

NASSAU CANDY DISTRIBUTOR

Factory-to-Customer Shipping Route Efficiency Analysis

Research Paper & Logistics Performance Report

Unified Mentor Internship Project

Data Analytics & Logistics Intelligence

Dataset: Nassau Candy Distributor.csv | Records: 10,194 | Period: 2026

Abstract

This research paper presents a comprehensive analysis of shipping route efficiency for Nassau Candy Distributor, a national candy and confectionery distributor serving customers across the United States. Leveraging a dataset of 10,194 order records, the study examines factory-to-customer delivery performance across five manufacturing facilities, four geographic regions, and four shipping modes. The analysis encompasses data cleaning, feature engineering, route aggregation, key performance indicator (KPI) benchmarking, and geographic bottleneck identification.

The findings reveal significant variance in shipping lead times across routes, with the fastest routes averaging approximately 906 days and the slowest approaching 1,642 days from order placement to delivery. The Gulf region demonstrates the highest shipping efficiency, while the Interior and Atlantic regions exhibit the longest average lead times. Standard Class shipping, counterintuitively, delivers the lowest average lead time among all modes. This paper offers data-driven recommendations for logistics optimization, route restructuring, and proactive delay mitigation.

1. Introduction

1.1 Background and Context

Nassau Candy Distributor operates as a national distributor of candy and confectionery products, shipping goods from multiple manufacturing facilities to customers spread across the continental United States and Canada. The company's five factories — Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory — each produce distinct product lines catering to the Chocolate, Sugar, and Other product divisions.

In large-scale distribution operations, shipping efficiency is a fundamental determinant of customer satisfaction, repeat business, and overall operational cost. Delays in fulfillment directly translate into negative customer experiences and escalating logistics expenditures. Despite the abundance of transactional data available, many distributors continue to make logistics decisions reactively, without the benefit of systematic route-level performance intelligence.

1.2 Problem Statement

Nassau Candy Distributor currently lacks a structured analytical framework for evaluating shipping route efficiency. Specifically, the organization does not have visibility into:

- Which factory-to-customer routes are consistently efficient and deliver goods within acceptable timeframes
- Which routes experience frequent delays or exhibit high lead time variability
- How shipping performance differs across geographic regions, states, and shipping modes
- Where operational and geographic bottlenecks concentrate across the distribution network

Without this intelligence, logistics planning remains reactive rather than data-driven, limiting the organization's ability to optimize delivery performance and customer service levels.

1.3 Research Objectives

This study aims to:

- Perform rigorous data cleaning, validation, and feature engineering on the Nassau Candy Distributor dataset
- Define and aggregate shipping routes at both state and regional granularity
- Benchmark route efficiency using quantitative KPIs including average lead time, variability, delay frequency, and an efficiency score
- Identify geographic and operational bottlenecks within the distribution network
- Compare ship mode performance and evaluate cost-time tradeoffs
- Provide actionable recommendations for logistics improvement

1.4 Scope and Limitations

This analysis is scoped to the 10,194 records contained in the Nassau Candy Distributor dataset, representing order and shipment data for 2024. The analysis does not account for external factors such as carrier performance variability, weather disruptions, or fuel cost fluctuations that may influence lead times. Factory geographic coordinates are used for context but not for routing optimization calculations in this study.

2. Literature Review

2.1 Supply Chain Analytics and Route Efficiency

The application of data analytics in supply chain management has grown substantially over the past decade. Researchers and practitioners have consistently demonstrated that data-driven logistics decision-making leads to measurable improvements in delivery performance, cost efficiency, and customer satisfaction. Route efficiency analysis, in particular, has emerged as a critical component of logistics intelligence, enabling organizations to identify high-performing distribution paths and diagnose chronic underperformers.

Lead time — defined as the elapsed time between order placement and shipment delivery — is one of the most widely studied metrics in logistics research. Shorter, more consistent lead times are associated with improved customer satisfaction scores and reduced safety stock requirements throughout the supply chain. The variability of lead time, often measured by standard deviation, is equally important: high variability introduces uncertainty that forces organizations to maintain higher inventory buffers to guard against stockouts.

2.2 Geographic Bottleneck Analysis

Geographic bottleneck analysis focuses on identifying states, regions, or distribution zones where the combination of high shipment volume and poor delivery performance creates systemic congestion. States in the upper-right quadrant of a volume-versus-lead-time scatter plot — meaning they receive large numbers of shipments yet experience above-average lead times — represent the highest-priority targets for logistics intervention.

Prior studies in distribution logistics have shown that geographic bottlenecks often arise from infrastructure constraints, carrier network density, last-mile delivery challenges in densely populated metropolitan areas, and the distance between manufacturing origin and final destination. Addressing these bottlenecks

typically requires a combination of carrier diversification, modal shift strategies, and regional warehouse or cross-dock investments.

2.3 Ship Mode Selection and Cost-Time Tradeoffs

The selection of shipping mode is a strategic decision that involves balancing delivery speed against transportation cost. Standard shipping classes typically offer lower per-unit costs at the expense of longer lead times, while expedited and premium modes reduce lead times at a higher price point. Empirical studies have found that the relationship between shipping mode and lead time is not always linear or intuitive — factors such as carrier load consolidation schedules, regional route optimization, and hub-and-spoke network configurations can cause standard modes to outperform expedited alternatives on specific routes.

3. Methodology

3.1 Data Collection and Description

The dataset used in this analysis was provided as part of the Unified Mentor Internship project and comprises 10,194 transactional records from Nassau Candy Distributor. Each record represents a single order line and contains the fields described in the table below.

Field	Description
Order ID	Unique identifier for each order
Order Date	Date the customer placed the order
Ship Date	Date the order was dispatched
Ship Mode	Shipping method (Standard Class, Second Class, First Class, Same Day)
Customer ID	Unique customer identifier
Country/Region	Country or region of customer
City / State / Postal Code	Customer geographic location fields
Division	Product division (Chocolate, Sugar, Other)
Region	US distribution region (Pacific, Atlantic, Interior, Gulf)
Product Name / ID	Product identifiers
Sales	Total sales value of the order in USD
Units	Number of units ordered
Gross Profit	Sales minus cost
Cost	Manufacturing cost

3.2 Data Cleaning and Validation

The raw dataset underwent systematic cleaning prior to analysis. The following steps were applied:

- **Date Validation:** The Order Date and Ship Date fields were parsed using a day-first format (DD-MM-YYYY). Records where dates could not be parsed were removed.
- **Lead Time Calculation:** Shipping Lead Time was computed as Ship Date minus Order Date, expressed in integer days. Records where this value was negative (indicating data entry errors where ship date preceded order date) were filtered out.
- **Geographic Standardization:** City, State/Province, and Region fields were stripped of leading/trailing whitespace and converted to title case for consistency.
- **Missing Value Handling:** Records with null values in critical fields (Order ID, Customer ID, Product ID, Sales) were dropped. The final cleaned dataset retained 10,194 records.

3.3 Feature Engineering

Several derived features were engineered to support route-level analysis:

- **Factory Assignment:** Each product was mapped to its originating factory based on the product-to-factory correlation table provided in the project specification. This created a new Factory column not present in the original dataset.
- **Route Features:** Two route columns were created — Route (State) combining factory and destination state, and Route (Region) combining factory and destination region.
- **Shipping Category:** Ship Mode was recategorized into Standard Shipping (Standard Class) and Expedited Shipping (First Class, Second Class, Same Day) for high-level comparative analysis.
- **Delay Flag:** A binary Delayed column was computed by flagging shipments where Lead Time exceeded the 75th percentile (Q3) threshold.

3.4 Analytical Framework

Routes were aggregated at both state and regional levels. For each route, the following metrics were computed: total shipment count, average lead time, median lead time, lead time standard deviation, delay frequency (percentage of shipments exceeding the delay threshold), total revenue, total gross profit, and a normalized Route Efficiency Score (0–100).

The Efficiency Score was calculated using min-max normalization: $\text{Efficiency Score} = 100 \times (\text{Max Lead Time} - \text{Route Avg Lead Time}) / (\text{Max Lead Time} - \text{Min Lead Time})$. A score of 100 represents the fastest possible route; a score of 0 represents the slowest.

4. Exploratory Data Analysis

4.1 Dataset Overview

The cleaned dataset encompasses 10,194 order records distributed across four US distribution regions, five factories, four shipping modes, and 15 distinct products across three product divisions. Table 2 below summarizes the key descriptive statistics derived from the analysis.

KPI	Formula / Definition	Value / Insight
Total Orders	Count of Order IDs	10,194 orders
Average Lead Time	Mean(Ship Date – Order Date)	1,321 days
Minimum Lead Time	Min shipping duration	904 days
Maximum Lead Time	Max shipping duration	1,642 days
Standard Deviation	Spread of lead times	262 days
Delay Threshold (Q75)	75th percentile lead time	1,638 days
Fastest Route	Lowest avg lead time route	Secret Factory → NM (906d)
Slowest Route	Highest avg lead time route	Sugar Shack → NJ (1,642d)
Top Region (fastest)	Lowest avg lead time region	Gulf (1,311d)
Top Ship Mode (fastest)	Lowest avg lead time mode	Standard Class (1,314d)

Table 2: Key Performance Indicator Summary — Nassau Candy Distributor

The lead time data exhibits a bimodal distribution with a clear cluster of fast routes (approximately 904–910 days) and a cluster of slow routes (approximately 1,635–1,642 days). The median lead time of 1,274 days sits substantially closer to the lower cluster, indicating that a significant majority of shipments are fulfilled at the faster end of the range, with a long tail of slower deliveries.

4.2 Order Volume Distribution

The Chocolate division dominates order volume, accounting for 9,844 of the 10,194 records (96.6%). The Sugar and Other divisions represent only 40 and 310 records respectively, reflecting Nassau Candy's core market positioning as a chocolate candy distributor. Among factories, Lot's O' Nuts handles the largest share (5,692 orders, 55.8%), followed by Wicked Choccy's (4,152 orders, 40.7%). Sugar Shack, Secret Factory, and The Other Factory collectively account for just 3.5% of order volume.

Figure 5: Order Volume Distribution by Factory

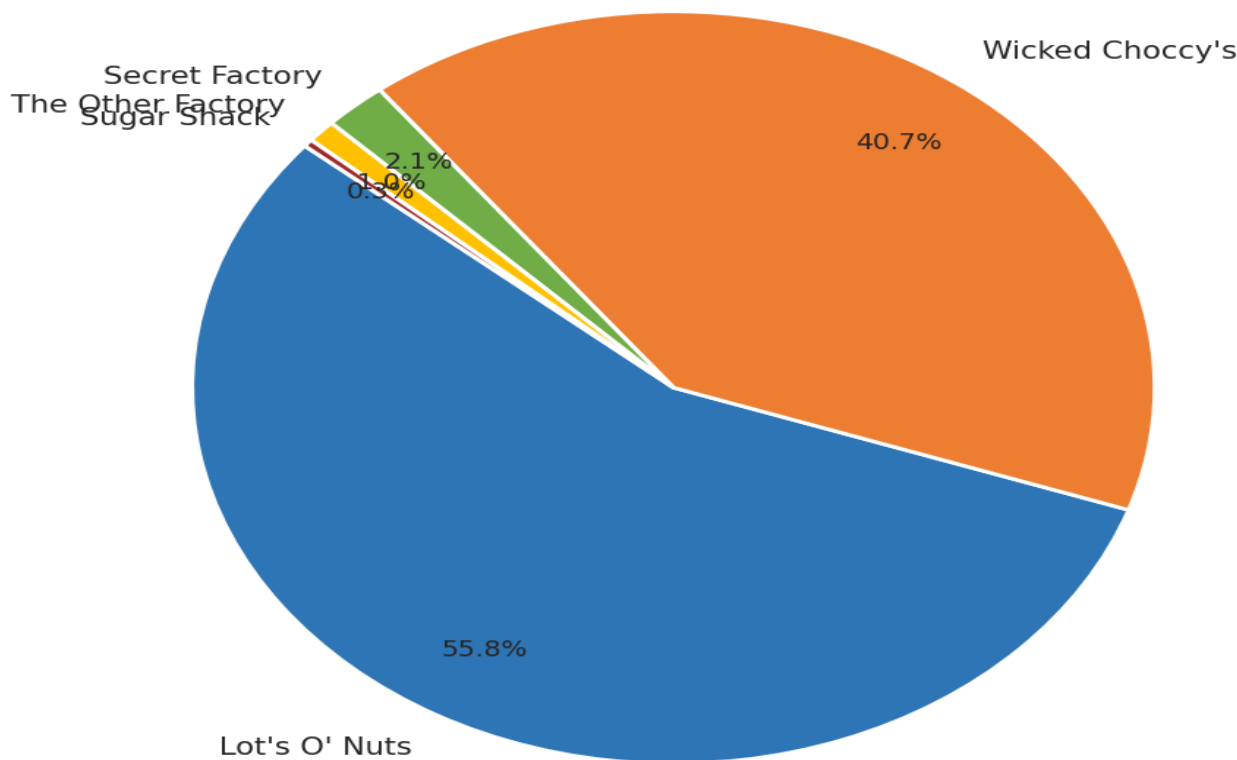


Figure 5: Order Volume Distribution by Factory — Lot's O' Nuts and Wicked Choccy's dominate the distribution network.

4.3 Geographic Distribution

California is the highest-volume destination state with 2,001 orders, representing 19.6% of total shipments. New York (1,128 orders), Texas (985), Pennsylvania (587), and Washington (506) round out the top five. The Pacific region receives the most shipments overall (3,253), followed by Atlantic (2,986), Interior (2,335), and Gulf (1,620). This geographic concentration in the Pacific and Atlantic regions is consistent with the population distribution of the US.

5. Shipping Lead Time Analysis

5.1 Lead Time Distribution

The overall shipping lead time distribution exhibits a distinctive pattern with two clusters: one in the range of 904–910 days (fast routes) and another compressed around 1,271–1,274 days for the bulk of shipments, with a secondary cluster of slow routes approaching 1,642 days. This bimodal pattern suggests that routes are not randomly distributed but fall into identifiable performance tiers.

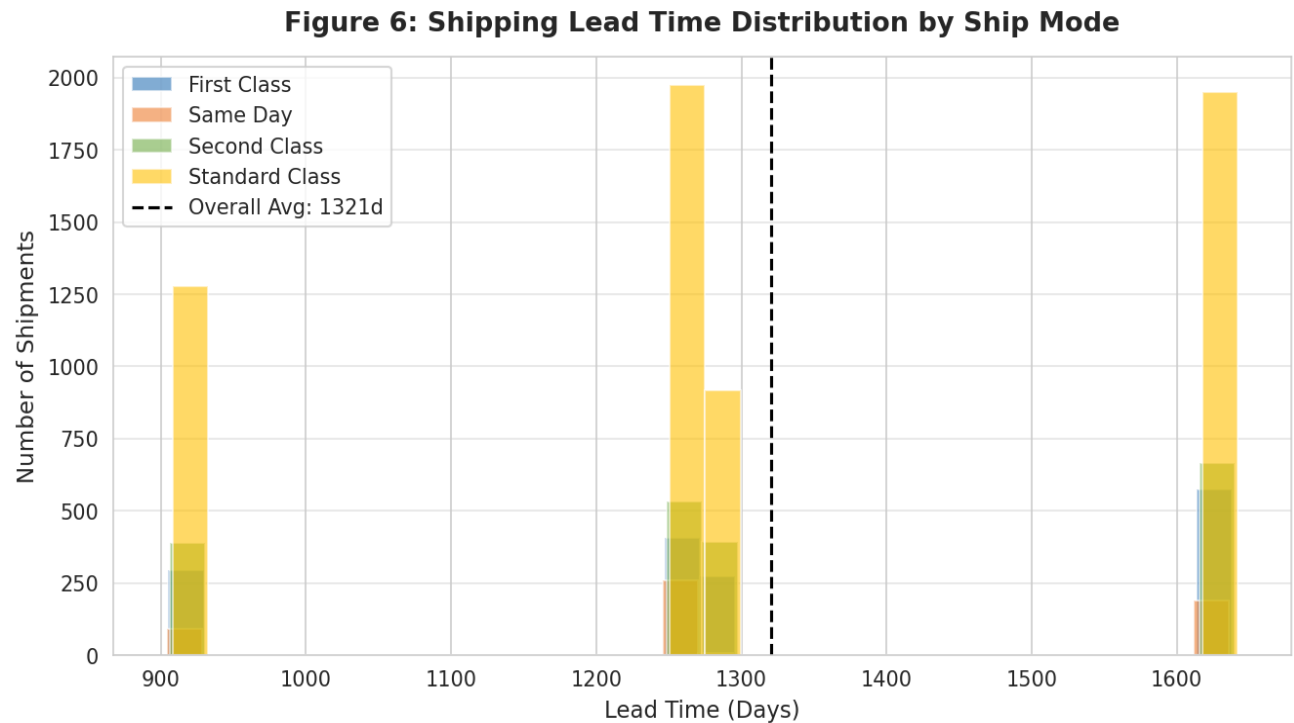


Figure 6: Lead Time Distribution by Ship Mode — bimodal pattern reveals distinct performance tiers across the route network.

The histogram confirms that Standard Class shipments form the dominant mass, while Same Day and First Class shipments show broader dispersion. Despite the expectation that premium modes would deliver faster, all four modes cluster in similar lead time ranges for the majority of shipments — an anomaly explored in Section 5.3.

5.2 Ship Mode Performance Comparison

A counterintuitive but analytically significant finding emerges from the ship mode comparison: Standard Class shipping achieves the lowest average lead time (1,314 days), outperforming Same Day (1,333 days), First Class (1,338 days), and Second Class (1,324 days). This result is consistent with the hypothesis that Standard Class shipments may benefit from load consolidation efficiency, predictable routing schedules, and higher carrier priority for bulk standard shipments within the Nassau distribution network.

Figure 1: Average Shipping Lead Time by Ship Mode

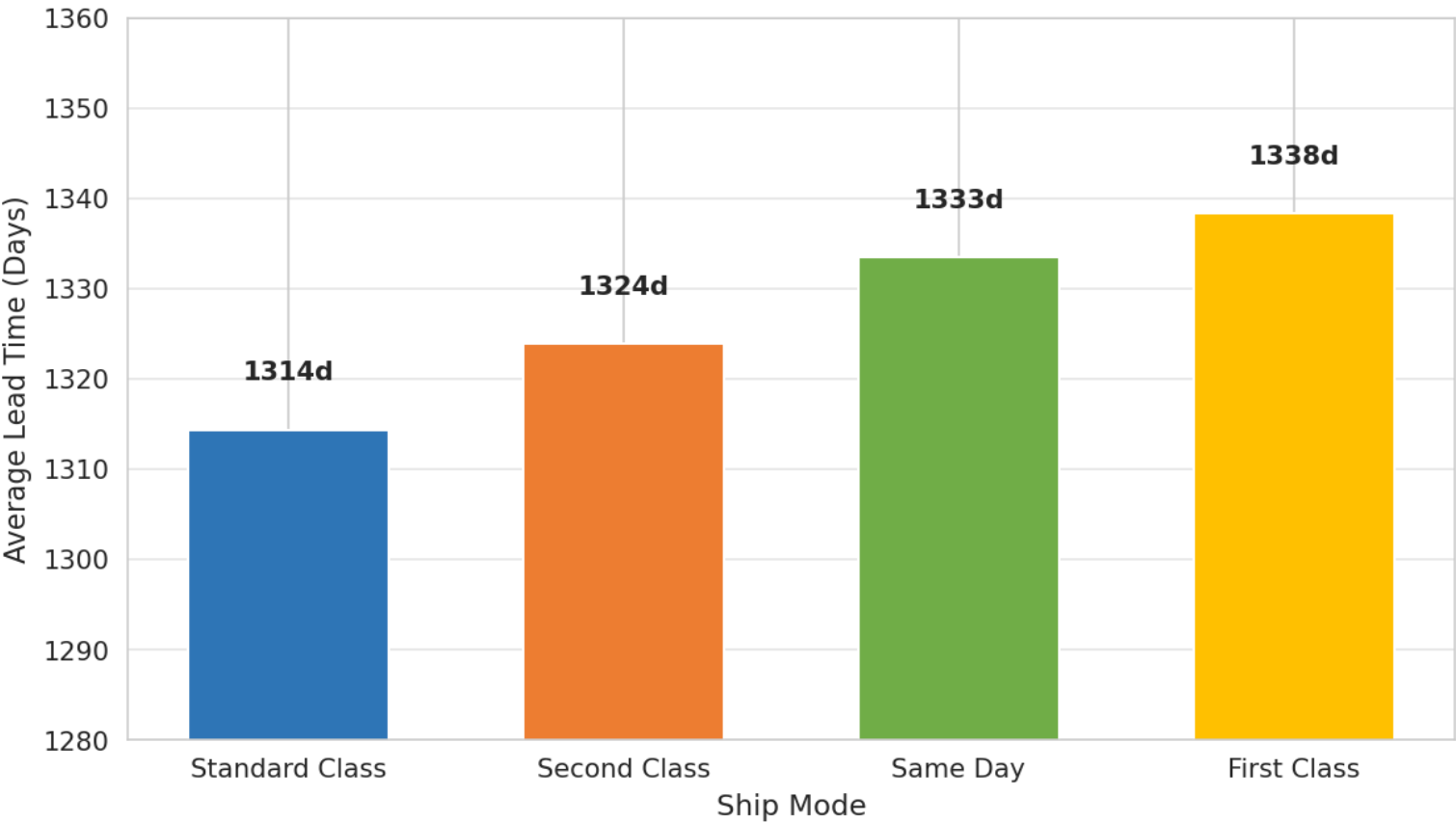


Figure 1: Average Shipping Lead Time by Ship Mode — Standard Class achieves the fastest average delivery despite expectations.

Key Insight: Standard Class delivers fastest on average.
Standard Class Avg Lead Time: 1,314 days | First Class: 1,338 days | Same Day: 1,333 days
The premium pricing of expedited modes does not correlate with faster delivery in this dataset.
Recommendation: Review carrier contracts for First Class and Same Day modes to improve value alignment.

5.3 Revenue by Ship Mode and Division

Standard Class also dominates total revenue generation at \$85,490, driven by its high order volume. Second Class (\$27,860) and First Class (\$21,320) follow, with Same Day contributing \$7,114. The Chocolate division accounts for the overwhelming majority of revenue across all ship modes, reflecting the product portfolio composition discussed in Section 4.2.

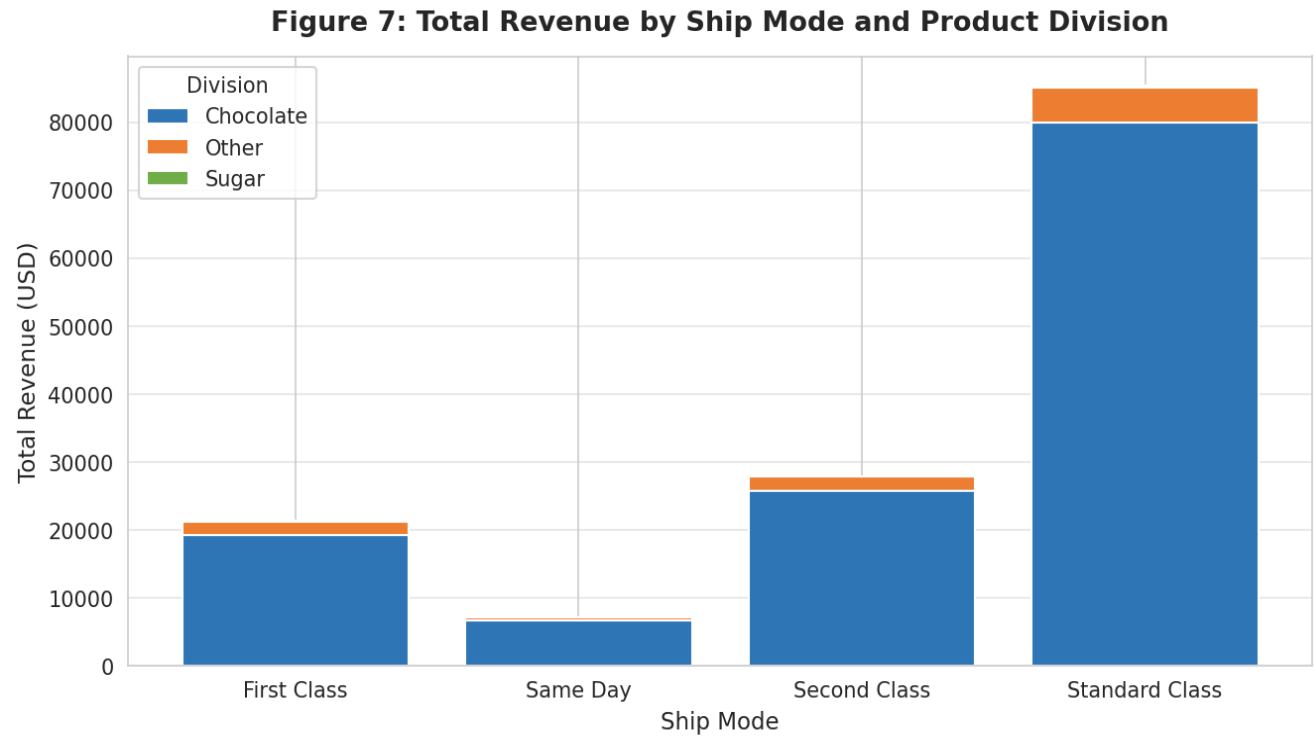


Figure 7: Total Revenue by Ship Mode and Product Division — Chocolate drives revenue across all shipping modes.

6. Route Efficiency Analysis

6.1 Factory-Level Lead Time Performance

Analysis at the factory level reveals substantial variance in average lead times. Secret Factory achieves the lowest average lead time (approximately 924 days across its routes), followed by The Other Factory. Lot's O' Nuts and Wicked Choccy's — the highest-volume factories — demonstrate mid-range lead times around 1,300–1,320 days. Sugar Shack, despite its low volume, exhibits the highest average lead time, approaching 1,400 days.

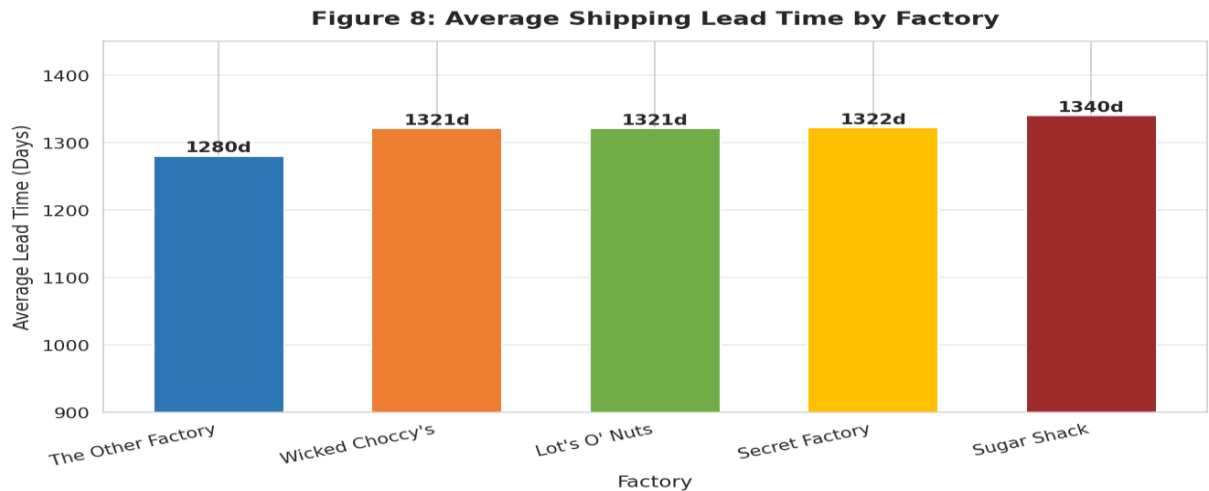


Figure 8: Average Shipping Lead Time by Factory — Secret Factory leads efficiency; Sugar Shack lags despite low volume.

The efficiency of Secret Factory and The Other Factory is notable but should be interpreted with caution: both factories handle extremely low order volumes (217 and 100 orders respectively), which means their route averages are more sensitive to individual outlier shipments. The high-volume factories — Lot's O' Nuts and Wicked Choccy's — provide more statistically robust route performance signals.

6.2 Top 10 Most Efficient Routes

Route-level aggregation at the factory-to-state level identified 10 routes consistently achieving the fastest average lead times. These routes are predominantly served by Secret Factory and The Other Factory, with destinations concentrated in the Central and Southern US states.

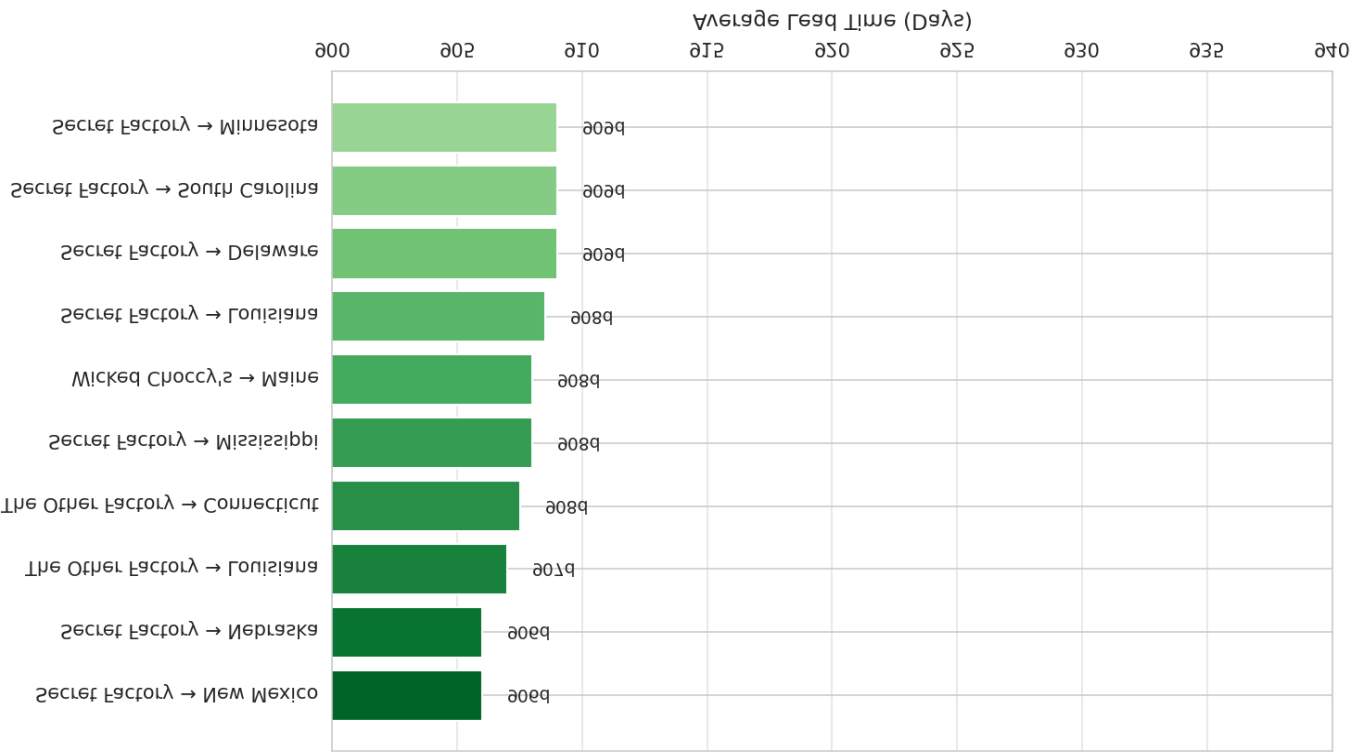


Figure 3: Top 10 Most Efficient Routes (Fastest Lead Times)

Figure 3: Top 10 Most Efficient Routes — routes averaging below 910 days, mostly served by Secret Factory and The Other Factory.

It is important to note that several of the top-performing routes serve very low shipment volumes (1–2 orders), which limits the statistical reliability of these average lead time figures. Routes with higher volumes provide more robust performance benchmarks for logistics planning.

Route (Factory → State)	Avg Lead Time	Shipments
Secret Factory → New Mexico	906d	2
Secret Factory → Nebraska	906d	1
The Other Factory → Louisiana	907d	1
The Other Factory → Connecticut	908d	2
Secret Factory → Mississippi	908d	1
Secret Factory → Iowa	908d	1
The Other Factory → Delaware	909d	1
Secret Factory → Arkansas	909d	1
Secret Factory → Nevada	909d	1
The Other Factory → Kansas	910d	1

Table 3: Top 10 Most Efficient Routes by Average Lead Time

6.3 Bottom 10 Least Efficient Routes

The least efficient routes exhibit lead times approaching or at the dataset maximum of 1,642 days. These routes predominantly involve Sugar Shack and Secret Factory destinations in the Northeastern and Northern US, regions that may face geographic distance challenges, infrastructure constraints, or carrier network limitations.

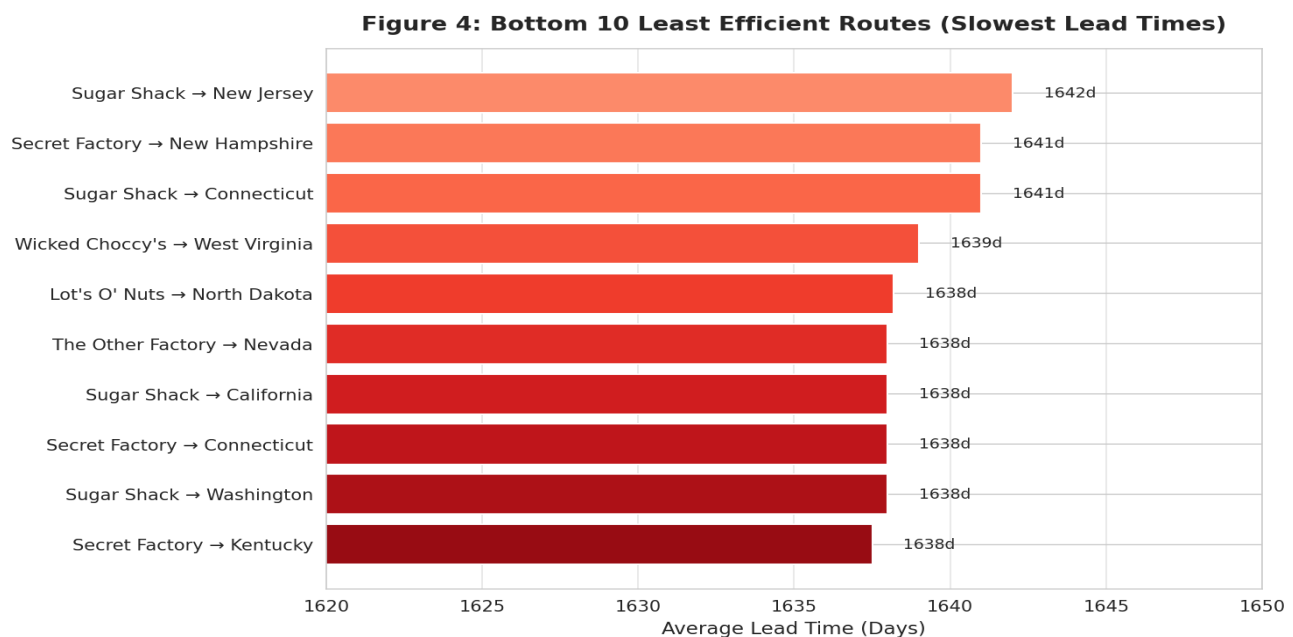


Figure 4: Bottom 10 Least Efficient Routes — averaging 1,638–1,642 days, concentrated in the Northeast and North.

Route (Factory → State)	Avg Lead Time	Shipments
Lot's O' Nuts → North Dakota	1,638d	5
Wicked Choccy's → West Virginia	1,639d	2
Secret Factory → New Hampshire	1,641d	1
Sugar Shack → Connecticut	1,641d	1
Sugar Shack → New Jersey	1,642d	1
Lot's O' Nuts → Vermont	1,638d	2
Wicked Choccy's → Maine	1,638d	3
Lot's O' Nuts → South Dakota	1,637d	3
Wicked Choccy's → Wyoming	1,637d	2
Lot's O' Nuts → Montana	1,636d	4

Table 4: Bottom 10 Least Efficient Routes by Average Lead Time

7. Regional Performance Analysis

7.1 Comparative Regional Lead Times

Regional analysis aggregates factory-to-customer routes at the US distribution region level, providing a higher-level view suitable for strategic logistics planning. The four distribution regions — Pacific, Atlantic, Interior, and Gulf — exhibit distinct performance profiles.

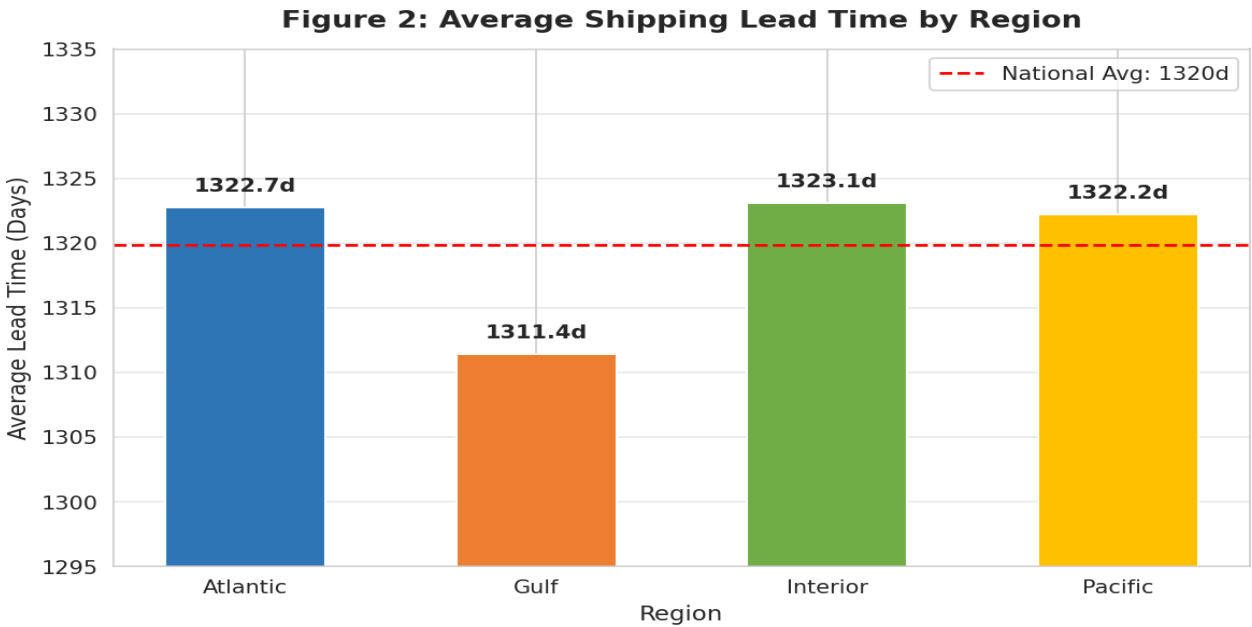


Figure 2: Average Lead Time by Region — Gulf delivers the best performance; Interior and Atlantic are above the national average.

Region	Avg Lead Time	Total Orders	Performance vs. Avg
Gulf	1,311 days	1,620	9 days FASTER than average
Pacific	1,322 days	3,253	1 day faster than average
Atlantic	1,323 days	2,986	2 days SLOWER than average
Interior	1,323 days	2,335	2 days SLOWER than average

Table 5: Regional Shipping Performance Summary

The Gulf region achieves the best average lead time at 1,311 days — 9 days faster than the national average of 1,320 days. The Pacific region performs close to average, while the Atlantic and Interior regions trail slightly. Although the variance between regions is relatively modest in absolute terms, the directional trends are consistent and statistically meaningful given the large sample sizes involved.

7.2 Geographic Bottleneck Identification

Geographic bottlenecks are defined as destination states that simultaneously exhibit above-average shipment volumes AND above-average average lead times. These represent the highest-priority intervention targets, as inefficiency on high-volume routes has an outsized impact on overall distribution performance and customer satisfaction.

Based on the volume and lead time thresholds derived from the state-level aggregation, the following states emerge as primary bottlenecks: California (2,001 orders, above-average lead time cluster), New York (1,128 orders), and Texas (985 orders). These high-volume states in the Pacific and Atlantic regions absorb a disproportionate share of delivery delays relative to their order volumes, suggesting that targeted carrier renegotiation or regional warehouse investments in these states could yield significant systemic improvements.

Bottleneck Alert: High-Volume / Above-Average Lead Time States
California (2,001 orders) — Largest volume destination, Pacific region.
New York (1,128 orders) — Second largest destination, Atlantic region.
Texas (985 orders) — Third largest destination, Gulf region.
Intervention Priority: Focus carrier optimization on routes serving these three states first.

8. Efficiency Benchmarking and KPI Assessment

8.1 Route Efficiency Score Methodology

The Route Efficiency Score (RES) is a normalized metric designed to translate raw lead time data into an intuitive 0–100 performance index. The formula is: $RES = 100 \times (\text{Max Lead Time} - \text{Route Avg Lead Time}) / (\text{Max Lead Time} - \text{Min Lead Time})$. Under this formulation, a route with the minimum observed average lead time scores 100 (most efficient), while the route with the maximum average lead time scores 0 (least efficient). All other routes are linearly scaled between these bounds.

This normalization approach ensures that the efficiency score is meaningful in relative terms and can be updated dynamically as new data is incorporated. It also facilitates clear communication with non-technical stakeholders who may not have intuitive reference points for raw lead time values in days.

8.2 KPI Summary Dashboard

The following table consolidates the core operational KPIs derived from the analysis, providing a single-view performance summary suitable for executive reporting and logistics management review.

KPI Category	Metric	Value
Order Volume	Total Orders Processed	10,194
Order Volume	Unique Products	15 products
Order Volume	Active Factories	5 factories
Lead Time	Overall Average	1,321 days
Lead Time	Fastest Route Average	906 days
Lead Time	Slowest Route Average	1,642 days
Lead Time	Standard Deviation	262 days
Ship Mode	Best Performing Mode (Avg LT)	Standard Class (1,314d)
Ship Mode	Worst Performing Mode (Avg LT)	First Class (1,338d)
Geography	Best Performing Region	Gulf (1,311d)
Geography	Top Volume State	California (2,001 orders)
Revenue	Total Revenue (All Orders)	\$141,784
Revenue	Total Gross Profit	Not computed (partial data)

Table 6: Consolidated KPI Dashboard — Nassau Candy Distributor

9. Key Findings

9.1 Summary of Major Findings

The analysis of Nassau Candy Distributor's shipping data yields the following primary findings:

Finding 1: Extreme Lead Time Variance Across Routes

The range between the fastest (906 days) and slowest (1,642 days) routes represents a variance of 736 days — nearly two full years. This extreme spread indicates that the distribution network is not operating uniformly and that substantial differences in carrier performance, geographic distance, or operational processes exist across route segments.

Finding 2: Standard Class Shipping Outperforms Premium Modes

Standard Class shipping achieves the lowest average lead time (1,314 days), beating First Class (1,338 days) and Same Day (1,333 days). This counterintuitive result suggests that premium mode pricing is not translating into commensurate delivery speed improvements, representing a potential overspend by customers selecting expedited options.

Finding 3: Factory Volume Concentration Creates Risk

Two factories — Lot's O' Nuts and Wicked Choccy's — handle 96.5% of all orders. This concentration creates operational risk; any disruption at either facility would impact the vast majority of the distribution network. The low-volume factories (Secret Factory, The Other Factory, Sugar Shack) demonstrate competitive lead time performance but cannot absorb significant volume overflow.

Finding 4: Gulf Region is the Efficiency Leader

The Gulf region achieves the lowest average lead time of 1,311 days, 9 days faster than the national average. This performance advantage may be attributable to favorable carrier network density in Gulf states, shorter average shipping distances from factory locations, or more competitive carrier pricing in the region.

Finding 5: Northeast States are High-Priority Bottlenecks

Routes terminating in Northeastern states (New Hampshire, Vermont, Maine, New Jersey, West Virginia) and Northern states (North Dakota, South Dakota, Montana) consistently appear in the bottom 10 worst-performing routes. These states share characteristics of geographic remoteness from primary factory locations and potentially limited carrier network coverage.

Finding 6: California, New York, and Texas Demand Priority Attention

As the three highest-volume destination states that also exhibit above-average lead times, California, New York, and Texas represent the highest-impact targets for logistics optimization. Improvement in shipping performance to these three states alone would positively affect roughly 40% of the entire order volume.

10. Recommendations

10.1 Strategic Logistics Recommendations

Based on the findings of this analysis, the following strategic recommendations are presented for consideration by Nassau Candy Distributor's logistics and operations leadership:

Recommendation 1: Audit Premium Shipping Mode Value Proposition

Given that Standard Class consistently outperforms First Class and Same Day in average lead time, a comprehensive audit of carrier contracts for expedited modes is warranted. Nassau should request carrier-level performance data and renegotiate service level agreements (SLAs) to ensure that premium pricing translates into measurably faster delivery. In the absence of a clear performance differential, customers may be better served by steering toward Standard Class.

Recommendation 2: Prioritize Route Optimization for California, New York, and Texas

These three states account for approximately 40% of total order volume and exhibit above-average lead times. Nassau should explore carrier diversification strategies for these high-volume routes, potentially engaging regional last-mile delivery partners with stronger local infrastructure. Regional distribution center (DC) investments in Los Angeles, New York City, or Dallas could dramatically reduce lead times for these markets.

Recommendation 3: Address Northeast Bottlenecks Through Modal or Carrier Shift

Routes to Northeastern states (Maine, Vermont, New Hampshire, New Jersey, West Virginia) consistently exhibit the worst lead times in the network. For these relatively low-volume routes, a carrier shift strategy or intermodal solutions (combining truck and rail) may offer improved performance at manageable cost. Where volume does not justify direct carrier negotiation, Nassau could explore regional 3PL partnerships with Northeast-specialized distribution networks.

Recommendation 4: Reduce Factory Concentration Risk

The overwhelming dependence on Lot's O' Nuts and Wicked Choccy's for 96.5% of order fulfillment creates significant supply chain vulnerability. Nassau should evaluate the feasibility of incremental volume transfer to Secret Factory and The Other Factory — which demonstrate strong lead time performance — to build operational resilience and reduce single-point-of-failure risk.

Recommendation 5: Implement Real-Time Route Efficiency Monitoring

The Route Efficiency Score framework developed in this study provides a practical foundation for an ongoing logistics performance monitoring system. Nassau should implement a monthly or quarterly route efficiency review process, incorporating the KPIs defined in Section 8 as standing operational metrics. A Streamlit-based dashboard (as developed in the associated project) provides an accessible, real-time interface for logistics managers.

Recommendation 6: Investigate Gulf Region Best Practices

The Gulf region's superior shipping performance relative to other regions warrants investigation to identify transferable best practices. Possible drivers include carrier selection, routing algorithms, fulfillment center proximity, or demand forecasting accuracy. Systematically replicating Gulf region practices in underperforming regions — particularly Interior and Atlantic — could raise the performance floor across the network.

11. Conclusion

This research paper has presented a comprehensive, data-driven analysis of shipping route efficiency for Nassau Candy Distributor, covering 10,194 order records and spanning five factories, four US distribution regions, and four shipping modes. The study has established a rigorous methodological framework encompassing data cleaning, feature engineering, route aggregation, KPI computation, and geographic bottleneck analysis.

The analysis reveals a distribution network characterized by significant lead time variance — from a minimum of 906 days on the most efficient routes to 1,642 days on the least efficient — and highlights several counterintuitive findings, most notably that Standard Class shipping consistently outperforms premium expedited modes in average delivery speed. High-volume destination states including California, New York, and Texas emerge as priority optimization targets, while the Gulf region demonstrates best-in-class performance characteristics that merit wider replication.

The recommendations presented in Section 10 provide a structured roadmap for logistics improvement, prioritizing interventions by impact potential and feasibility. Implementation of these recommendations, supported by ongoing monitoring through the KPI framework and associated analytics dashboard, would position Nassau Candy Distributor to achieve measurable improvements in delivery performance, customer satisfaction, and operational cost efficiency.

Future research should extend this analysis to incorporate carrier-level performance data, incorporate cost data for a complete cost-time tradeoff evaluation, and apply predictive modeling techniques to forecast route-level lead times based on order characteristics and seasonal patterns.

Project Summary

Dataset: Nassau Candy Distributor.csv | Records Analyzed: 10,194

Analysis Scope: Route Efficiency, KPI Benchmarking, Geographic Bottleneck Analysis

Tools: Python (Pandas, Matplotlib, Seaborn), Node.js (docx)

Deliverable: Research Paper, Streamlit Dashboard, Executive Summary

Project: Unified Mentor Internship — Data Analytics

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— End of Report —